

DATE:18.08.26

EX:NO:17

Feed Forward Neural Network

AIM

To implement a Feed Forward Neural Network and display its training error graphically.

ALGORITHM

1. Create input and target data.
2. Initialize weights.
3. Perform forward propagation.
4. Calculate the error.
5. Update the weights.
6. Repeat the training process.
7. Plot the training error using Matplotlib.

CODE

```
import matplotlib.pyplot as plt
```

```
import math
```

```
import random
```

```
# Input data
```

```
X = [
```

```
    [0, 0],
```

```
    [0, 1],
```

```
    [1, 0],
```

```
    [1, 1]
```

```
]
```

```
# Target output
```

```
Y = [0, 1, 1, 0]
```

```
random.seed(1)
```

```
# Initial weights
```

```
w1 = random.uniform(-1, 1)
```

```
w2 = random.uniform(-1, 1)
```

```
w3 = random.uniform(-1, 1)
```

```
w4 = random.uniform(-1, 1)
```

```
learning_rate = 0.5
```

```
errors = []
```

```
def sigmoid(x):
```

```
    return 1 / (1 + math.exp(-x))
```

```
# Training
```

```
for epoch in range(5000):
```

```
    total_error = 0
```

```
    for i in range(len(X)):
```

```
x1 = X[i][0]
x2 = X[i][1]
target = Y[i]

# Hidden layer
h1 = sigmoid(x1 * w1 + x2 * w2)
h2 = sigmoid(x1 * w3 + x2 * w4)

# Output layer
output = sigmoid(h1 + h2)

# Error
error = target - output
total_error += error * error

# Simple weight update
w1 += learning_rate * error * x1
w2 += learning_rate * error * x2
w3 += learning_rate * error * x1
w4 += learning_rate * error * x2

errors.append(total_error / 4)
```

```
# Display predictions

print("Feed Forward Neural Network")

print("-----")

for i in range(len(X)):

    x1 = X[i][0]
    x2 = X[i][1]

    h1 = sigmoid(x1 * w1 + x2 * w2)
    h2 = sigmoid(x1 * w3 + x2 * w4)

    output = sigmoid(h1 + h2)

    prediction = 1 if output >= 0.5 else 0

    print(X[i], "=>", prediction)


# Plot graph

plt.figure(figsize=(8, 5))

plt.plot(errors)
```

```
plt.title("Feed Forward Neural Network Training")
```

```
plt.xlabel("Epoch")
```

```
plt.ylabel("Mean Squared Error")
```

```
plt.grid(True)
```

```
plt.show()
```

```
Feed Forward Neural Network
-----
... [0, 0] => 1
    [0, 1] => 1
    [1, 0] => 1
    [1, 1] => 1
```

